Which Isolates Should the Laboratory Test? Active Learning, Drift and Calibration for MALDI-TOF Antimicrobial Resistance Prediction

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Abstract

Machine learning models that predict antibiotic resistance from MALDI-TOF mass spectra lose accuracy as bacterial populations drift, so they must be retrained on freshly phenotyped isolates. Phenotyping is the expensive step, which raises a practical question that has not been studied for this modality: given a fixed number of susceptibility tests per year, *which* arriving isolates should the laboratory test? We report 2 859 retraining simulations on DRIAMS covering three organism and antibiotic pairs, four hospitals, label budgets from 40 to 800 and ten random seeds, comparing random selection against uncertainty, diversity and hybrid acquisition, against conformal safety routing, against pseudo labelling, and against recalibration of the deployed decision rule. Three findings stand out. First, **no acquisition strategy produced a better model than random sampling** in any of 45 preplanned comparisons, and the prespecified primary endpoint, the number of labels needed to recover 95 % of the full retraining gain, was never reached within 800 labels. Second, the choice nonetheless matters for patients: because a tested isolate receives a laboratory answer rather than a model answer, acquiring the uncertain isolates reduced the resistant isolates wrongly reported susceptible during the year by 33 to 73 % ($p < 0.01$ for all three pairs), with an effect on the opposite error that depends on the organism. Third, the safety machinery silently expires: a decision threshold calibrated for 90 % sensitivity delivered 73 to 80 % two years later, and a conformal rule promising at most 10 % miscoverage on resistant isolates delivered 14 to 19 %. Recalibrating on freshly acquired labels restored the sensitivity target but roughly doubled the opposite error, exposing the true operating cost of that promise at AUROC 0.75 to 0.85. A preregistered search over 16 method combinations, run on a development split that never touches the test year, reproduced the same picture and found no combination improving both error types beyond uncertainty acquisition itself. We conclude that label efficiency is the wrong objective for this problem, and that acquisition policies should be reported with counts of both error types at deployment and with calibration treated as a maintained component rather than a property of the training run.

Keywords: Antimicrobial resistance, MALDI-TOF mass spectrometry, Active learning, Dataset shift, Conformal prediction, Model calibration, Clinical decision support

1 Introduction

Antimicrobial resistance is established in routine practice by phenotypic susceptibility testing, which takes a day or more after the organism has been identified. Identification itself is nearly immediate, because clinical laboratories already acquire a MALDI-TOF mass spectrum of every isolate. Weis et al. [1] showed that those same spectra carry a usable resistance signal and released DRIAMS, a four hospital collection of spectra paired with laboratory susceptibility labels. A model that reads the spectrum can therefore offer a provisional resistance call hours before the phenotype is available, and reported areas under the receiver operating characteristic curve of approximately 0.80, 0.74 and 0.74 for *Escherichia coli*, *Klebsiella pneumoniae* and *Staphylococcus aureus* on clinically important antibiotics.

Such a model is not a static artefact. Bacterial populations, laboratory workflows and instruments change, and the same study documented that predictive accuracy decays over time. Keeping the model useful therefore requires retraining on recently phenotyped isolates, and phenotyping is exactly the cost the model was intended to avoid. A laboratory able to afford, for example, four hundred additional

susceptibility tests per year faces a concrete operational choice: *which* four hundred of the thousands of arriving isolates should be tested?

This is the classical pool based active learning question [2], and the natural hypothesis is that an informative subset buys more accuracy per test than a random subset. We set out to measure how much. Across 2 859 simulated retraining runs the answer is that no measurable advantage exists in this setting. The negative result is worth reporting on its own, because the opposite is commonly assumed in applied work. It is not, however, the whole story, and the remainder of this paper is concerned with what the negative result conceals.

The acquisition rule determines not only what the model learns, but also *who receives a laboratory answer instead of a model answer*. An isolate sent for testing is, from the patient's point of view, protected: its true phenotype becomes known before treatment decisions are finalised. An isolate that is not tested is reported with the model's call, errors included. Acquisition policy is therefore a patient safety decision taken during the year, largely independent of what it does for next year's model, and on that axis the strategies differ sharply.

A third thread runs through the results. Both the decision threshold and the conformal prediction sets used in this setting are calibrated on historical data, and both make explicit numerical promises: 90 % sensitivity to resistance in the first case, at most 10 % miscoverage per class in the second. Under drift both promises failed quietly while the headline discrimination metric remained healthy. We quantify the failure, repair it by recalibrating on newly acquired labels, and show that the repair exposes an operating cost which the broken calibration had been concealing.

The main contributions of this work are summarised as follows.

- The first study, to our knowledge, of **active isolate acquisition for the maintenance of MALDI-TOF resistance models**: six experiments and 2 859 simulated retraining runs over three organism and antibiotic pairs and four hospitals, with full seed replication and paired statistical testing.
- A clearly stated **negative result on label efficiency**, reported together with the statistical power that supports it, alongside a **positive result at deployment** concerning the errors that actually reach patients.
- Quantification of the **decay of threshold and conformal calibration under temporal drift**, and a measurement of what restoring the sensitivity promise costs in terms of the opposite error.
- A **preregistered combination search** over sixteen method variants, conducted on a development split that never touches the test year, which reproduces the conclusions and eliminates the obvious remedies.
- An explicit account of a **design flaw in our own semisupervised experiment**, reported rather than quietly discarded, so that the corresponding question is left open rather than answered incorrectly.

The remainder of the paper is organised as follows. Section 2 reviews related work on resistance prediction from mass spectra, active learning and calibration under shift. Section 3 formalises the maintenance loop, the error taxonomy and the acquisition strategies. Section 4 describes the data, the experimental matrix and the statistical protocol. Section 5 reports results. Section 6 discusses implications, Section 7 states limitations and Section 8 concludes.

2 Related Work

This section summarises progress in resistance prediction from mass spectra, uncertainty quantification for that modality, active learning and its behaviour under distribution shift, and calibration maintenance, and identifies the gap that motivates the present study.

2.1 Resistance Prediction from MALDI-TOF Spectra

DRIAMS established the task, the standard preprocessing and the reference baselines, assembling more than 300 000 spectra with laboratory confirmed resistance profiles from four Swiss diagnostic laboratories [1]. Subsequent work has extended the approach to further organisms and improved spectral representations. Temporal decay of these models is documented within the same line of work, so demonstrating decay is not a contribution of this paper; it is the premise on which the study rests, and we verify it again under our own protocol in Section 5.1.

2.2 Uncertainty Quantification for This Modality

Classifiers intended for clinical microbiology must be able to decline a prediction when they cannot make one reliably. Corvelo Benz et al. [3] applied conformal prediction to MALDI-TOF resistance classification, obtaining prediction sets with distribution free statistical guarantees, and earlier work introduced kernels tailored to mass spectra with well calibrated uncertainty estimates [4]. Our use of conformal sets as an *acquisition* rule, that is, to test precisely those isolates the model cannot resolve, builds directly on that work. Our contribution is the observation that the guarantee does not survive a year of drift unless it is recalibrated, and a measurement of the size of the violation.

2.3 Active Learning and Query Strategies

Pool based active learning iteratively requests expert annotation for the most informative unlabelled samples [2]. Uncertainty sampling [5], core set and k-center diversity [6], and hybrids of the two are the standard families. Importance weighted [7], bandit selected [8] and cost sensitive [9] variants address sampling bias, strategy selection and asymmetric misclassification cost respectively. That random acquisition is a stubborn baseline is itself a recurring observation in benchmark studies, and deliberately injecting randomness into uncertainty sampling is a known stabiliser [10]. In medical imaging, active learning has reduced annotation burden substantially, but the evaluation is almost always label efficiency on a held out set rather than the errors incurred while the loop runs.

2.4 Active Learning under Distribution Shift

Malware detection faces the same structure as our problem, namely a deployed classifier, continuous arrival of new samples, a bounded analyst budget and continuous drift, and label efficient update policies [11] together with the decay of conformal evaluation under drift [12] are both documented there. The active selective prediction setting combines acquisition with the option to abstain and transfer to a human [13]. None of this literature addresses which bacterial isolates a clinical laboratory should phenotype, and to our knowledge none of it uses the random share of an active learning budget as the calibration set for the deployed decision rule.

2.5 Calibration under Shift

Weighted conformal prediction restores coverage under covariate shift when a density ratio can be estimated [14], and adaptive conformal inference tracks shift online by adjusting the target level from observed feedback [15]. Both are implemented as candidate remedies in the combination search of Section 5.7. The underlying split conformal framework is standard [16].

3 Problem Setting and Method

3.1 The Maintenance Loop

We simulate the loop a laboratory would actually run, shown in Fig. 1. A model trained on historical isolates is in service. New isolates arrive through the year in $R = 4$ rounds. In round r, the model currently in service scores every arriving isolate; an acquisition rule spends b/R of a fixed annual budget b on susceptibility tests; the tested isolates return true labels, and the model, its decision threshold and its conformal bounds are refitted on everything acquired so far. Isolates that were not tested receive the model's call, and any error there is an error that a clinician may act on. At the end of the year the model is evaluated on the following year's isolates, which are never visible during the loop.

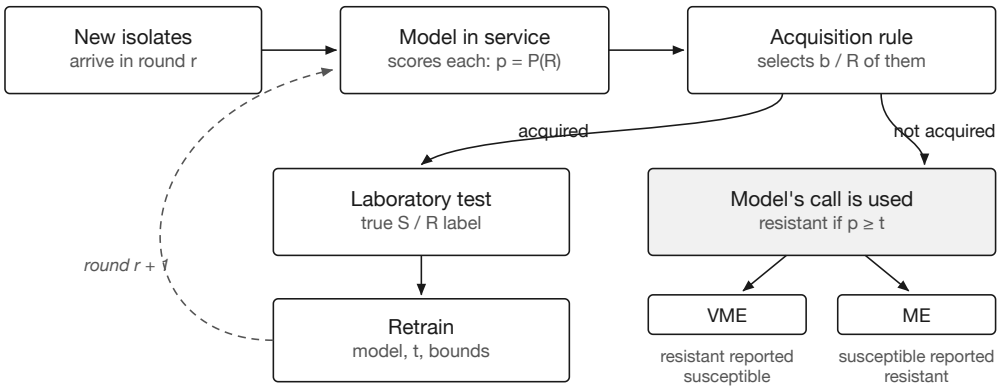


Fig. 1: The loop that every run simulates. The acquisition rule determines both what the model learns next (upper path) and which patients receive a laboratory answer rather than a model answer (lower path). Errors on the lower path are counted as they occur, not only at the end of the year. VME and ME are defined in Section 3.2.

3.2 Error Taxonomy and Reported Quantities

Two error types are distinguished throughout, following the terminology used to evaluate susceptibility testing devices [17]. A **very major error** (VME) is a resistant isolate reported as susceptible: the patient may receive an antibiotic that does not work. A **major error** (ME) is a susceptible isolate reported as resistant: the patient may receive an unnecessarily broad antibiotic, which carries its own harm and cost. The two are not interchangeable, and a method that lowers one while raising the other has not improved safety; it has moved along a tradeoff.

Both error types are reported in two distinct views, and the distinction is central to this paper. The *predictive* view gives rates on the following year's isolates and measures how good the retrained model is. The *operational* view gives counts among the untested isolates of the current year and measures how many patients were actually affected while the loop was running. Formally, writing U_r for the isolates of round r that the rule did not acquire, the operational very major error count is

$$\mathrm{VME}_{\mathrm{op}} = \Sigma_{\mathrm{r}=1..\mathrm{R}} \left| \left\{ \mathrm{i} \in \mathrm{U}_{\mathrm{r}} : \mathrm{y}_{\mathrm{i}} = \mathrm{R} \text{ and } \mathrm{p}_{\mathrm{i}} < \mathrm{t}_{\mathrm{r}} \right\} \right|,$$

where y_i is the true phenotype, p_i the probability assigned by the model in service when isolate i arrived, and t_r the threshold in force at that time; $\mathrm{ME}_{\mathrm{op}}$ is defined symmetrically. Because the rule and the model both change between rounds, these counts cannot be recovered from metrics computed at the end of the year, and they are the quantities on which acquisition strategies separate.

3.3 Acquisition Strategies

Let p be the model's probability of resistance for an arriving isolate and t the current decision threshold. Each round the rule selects b/R isolates from that round's arrivals.

- **Random.** Uniform selection without replacement; the control condition.
- **Uncertainty.** Smallest $|p - t|$, that is, the isolates whose call is closest to flipping [5].
- **Diversity.** Greedy k-center selection in a 50-dimensional principal component space, maximising coverage of spectrum space relative to what is already labelled [6].
- **Hybrid.** The $4b$ most uncertain isolates are clustered by k-means into b groups and the isolate nearest each centroid is acquired.
- **Conformal routing.** Only isolates whose per class prediction set is ambiguous, that is $\{\mathrm{S}, \mathrm{R}\}$ or empty, are acquired [3]. Reported both capped at the budget and uncapped.
- **Bounds.** *No retraining* (budget zero) and *full retraining* (every arriving isolate labelled) bracket what any rule can achieve.

3.4 Decision Threshold and Conformal Bounds

The operating threshold is chosen as the largest t achieving a target sensitivity $\tau = 0.90$ to resistance on held out data, which is the quantity a laboratory would specify. Mondrian (per class) split conformal prediction supplies prediction sets at $\alpha = 0.10$, calibrated separately on the susceptible and resistant validation isolates so that the minority class receives its own guarantee [16]. In the baseline configuration both are fitted once on the 2015 and 2016 validation split and then left untouched, which is what a deployed system does by default; Section 5.5 reports the consequence of refitting them each round on freshly acquired labels.

4 Experimental Setup

4.1 Dataset and Splits

All experiments use DRIAMS [1]. Each isolate is represented by one MALDI-TOF spectrum binned into 6 000 intensity features and paired with a laboratory susceptibility label; intermediate and resistant are both treated as resistant, following the reference implementation released with the dataset. We study the three organism and antibiotic pairs with sufficient labelled isolates for repeated simulation.

The temporal protocol, shown in Fig. 2, trains on 2015 and 2016 isolates from hospital DRIAMS-A, holding out 20 % as the validation split used for the threshold and the conformal bounds, spends the label budget on 2017 isolates arriving in four rounds, and evaluates on 2018. DRIAMS does not record acquisition dates, so the four arrival rounds are a fixed pseudo random partition of the 2017 isolates derived from a hash of the isolate identifier. This reproduces the structure of a budget spent over time but *cannot* reproduce drift within the year, a limitation discussed in Section 7. The transfer protocol moves the model trained on 2015 and 2016 to hospitals B, C and D and allows each to acquire local labels from its own 2018 isolates, holding out a stratified test split at each site.

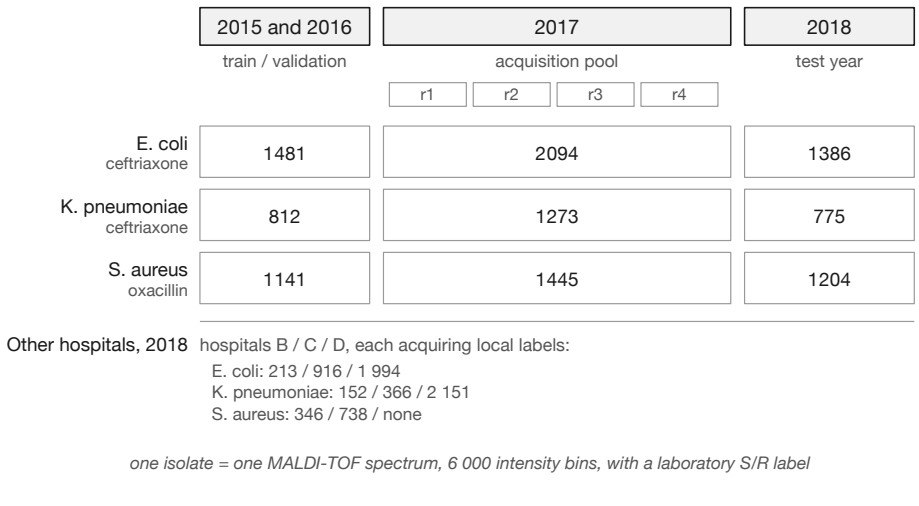


Fig. 2: Data and splits. Counts are isolates per organism and antibiotic pair. The 2018 isolates of hospital A are reserved for final evaluation, and the development split used for the combination search of Section 5.7 never touches them. *S. aureus* has no labelled oxacillin isolates at hospital D, so that combination is not run.

4.2 Experimental Matrix

Six experiments were executed, summarised in Table 1. Every configuration is replicated across seeds, and the data split, hyperparameters and random seed are held fixed within a comparison so that differences can be attributed to the acquisition rule and the calibration policy alone.

Table 1: Experimental matrix. Budgets are total labels acquired over the year, divided evenly across the four arrival rounds. Runs include the no retraining and full retraining bounds where applicable. One E5 run was lost in transfer, giving 239 of 240.

Id	Question	Acquisition rules	Budgets	Seeds	Runs
E1	Label efficiency and acquisition family	random, uncertainty, diversity, hybrid	40 to 800	10	660
E2	Value of unlabelled spectra	the above + pseudo labelling	100 to 800	10	390
E3	Conformal routing as an acquisition rule	routing, routing (uncapped)	40 to 800	10	180
E4	Adoption at a different hospital	random, uncertainty, diversity, hybrid	25 to 400	5	880
E5	Recalibration on acquired labels	random, uncertainty	100 to 800	10	239
D1	Preregistered combination search	16 combinations (Section 5.7)	100, 300	5	510
Total					**2 859**

4.3 Protocol Constants

Table 2 lists the quantities held constant across all runs. Hyperparameters were tuned once per organism and antibiotic pair on the training years by cross validation and then frozen, so that every acquisition rule is compared at equal model capacity.

Table 2: Protocol constants, held fixed across every run.

Parameter	Value
Organism and antibiotic pairs	*E. coli* with ceftriaxone, *K. pneumoniae* with ceftriaxone, *S. aureus* with oxacillin
Features	6 000 binned spectral intensities
Label convention	intermediate and resistant both treated as resistant
Training years / test year	2015 and 2016 (80 % train, 20 % validation) / 2018
Acquisition pool	2017 isolates, divided into $R = 4$ arrival rounds
Classifier	gradient boosted trees (XGBoost), tuned once per pair and then frozen
Target sensitivity τ	0.90, used to set the decision threshold
Conformal level α	0.10, per class (Mondrian) split conformal
Embedding for diversity	50 principal components fitted on the training years
Pseudo label cutoffs (E2)	$p \leq 0.05$ susceptible, $p \geq 0.95$ resistant, weight 0.5
Recalibration share (E5)	25 % of acquired labels withheld from training
Seeds	10 (5 for transfer and for the development search)
Statistical test	paired Wilcoxon signed rank over seeds, Holm correction within each pair

4.4 Evaluation Metrics and Statistical Analysis

Model quality is reported as AUROC on the following year's isolates, with area under the precision and recall curve and balanced accuracy as secondary measures. Safety is reported in the operational view of Section 3.2, as counts of both error types among untested isolates. Calibration is reported as realised sensitivity against the 0.90 target and as per class conformal coverage against the 0.90 guarantee. The prespecified primary endpoint was the number of labels required to recover 95 % of the AUROC gap between the two bounds, estimated by bootstrap over seeds. Strategies are compared with random acquisition by a paired Wilcoxon signed rank test over matched seeds, with the Holm procedure applied within each organism and antibiotic pair to control the family wise error rate at 0.05.

5 Results

5.1 Premise: Temporal Decay and the Value of Retraining

A model frozen after 2016 loses ground on 2018 isolates which full retraining on the 2017 labels recovers: AUROC rises by 0.044, 0.046 and 0.051 for the three pairs (Fig. 3). The gap is modest but entirely consistent across seeds, and it is the gap that every acquisition strategy competes to close cheaply. All subsequent results are expressed relative to these two bounds.

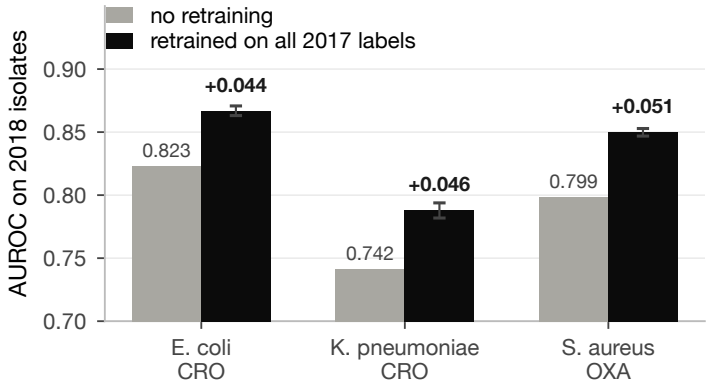


Fig. 3: Retraining recovers accuracy lost to drift. Bars show mean AUROC on 2018 isolates over ten seeds; whiskers are 95 % confidence intervals. Labels above the right hand bars give the gain over not retraining at all.

5.2 Primary Outcome: No Strategy Outperformed Random

Fig. 4 shows the label efficiency curves and Fig. 5 the paired differences against random acquisition. Across 45 preplanned comparisons, that is three pairs by five budgets by three strategies, **no strategy was significantly better than random after Holm correction**. The only significant results run in the opposite direction: uncertainty sampling was worse than random for *K. pneumoniae* at 800 labels (-0.029, Holm $p = 0.029$) and for *S. aureus* at the same budget (-0.013, raw $p = 0.006$, Holm $p = 0.088$). The prespecified primary endpoint was never reached: no strategy recovered 95 % of the retraining gap within 800 labels, and random acquisition itself recovered only 66 to 77 % at that budget. Values at the final budget are given in Table 3.

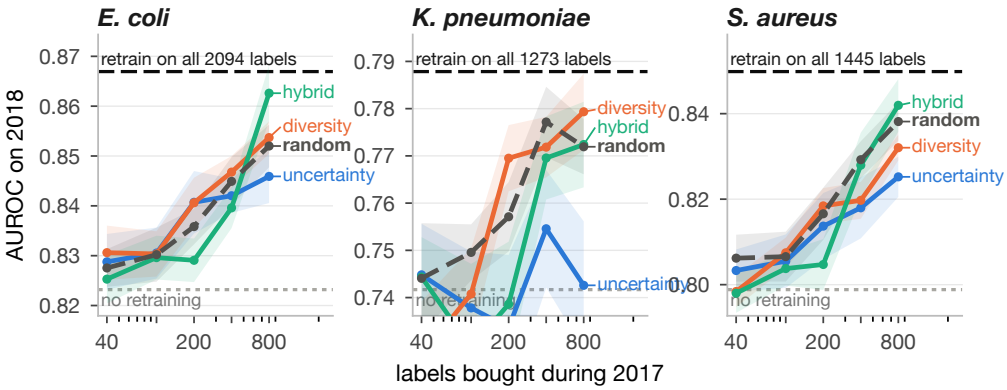


Fig. 4: Label efficiency curves. Mean AUROC on 2018 isolates over ten seeds, with shaded 95 % confidence intervals, against a logarithmic budget axis. The dashed upper line is full retraining, which consumes 1 273 to 2 094 labels, and the dotted lower line is no retraining. All four acquisition rules lie within one another's confidence bands at every budget.

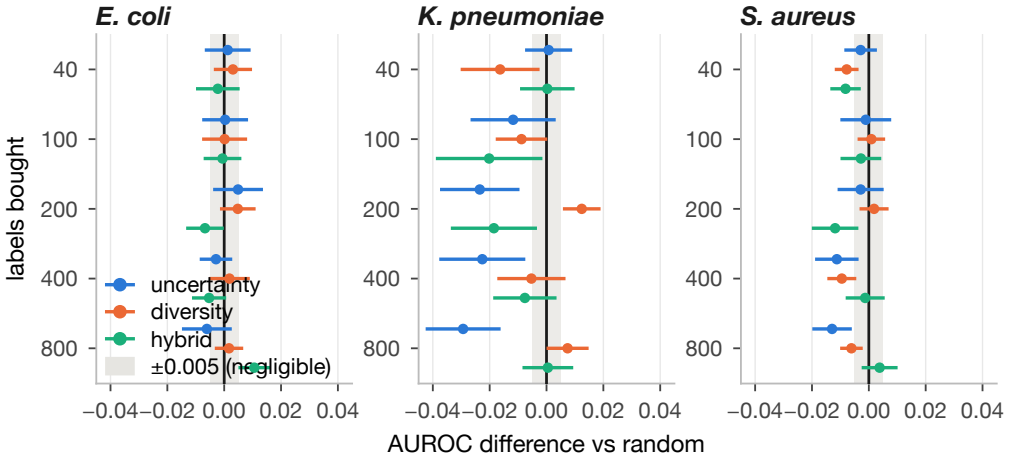


Fig. 5: Paired differences against random acquisition, by pair and budget. Points are mean AUROC differences over ten matched seeds with 95 % confidence intervals; the shaded band marks the difference of ±0.005 prespecified as negligible. No interval excludes zero in a direction favouring the strategy once Holm correction is applied.

Table 3: AUROC on 2018 isolates at the largest budget of 800 labels, mean over ten seeds, with the two bounds. Confidence intervals at 95 % are at most 0.014 throughout and are drawn in Figs. 4 and 5. The best active learning entry per row is set in bold; no difference from random is statistically significant.

Pair	No retraining	Random	Uncertainty	Diversity	Hybrid	Full retraining
E. coli / CRO	0.823	0.852	0.846	0.854	**0.863**	0.867
K. pneumoniae / CRO	0.742	0.772	0.743	**0.779**	0.772	0.788
S. aureus / OXA	0.799	0.838	0.825	0.832	**0.842**	0.850

The design is paired over seeds, which makes it sensitive: with ten seeds it detects an AUROC difference of approximately 0.008 at 80 % power for the observed variances. Large gains from these strategies can therefore be excluded in this setting, though gains of any size cannot. The practical reading is that a laboratory acquiring isolates at random gives up nothing measurable in model quality.

5.3 Acquisition Determines Which Patients Are Exposed

The picture changes once the measurement moves from next year's model to this year's patients. Because uncertainty sampling spends the budget precisely where the model's call is least trustworthy, the isolates it sends to the laboratory are disproportionately those the model would have got wrong, and those isolates then receive a true phenotype instead. Summed over the year at 800 labels, uncertainty acquisition reduced the resistant isolates wrongly reported susceptible by 33, 73 and 51 % relative to random (Fig. 6; all $p < 0.01$, paired Wilcoxon over seeds).

This benefit must be reported together with its cost on the opposite error, and here the species differ (Table 4). Major errors fell by 25 % for *E. coli*, were unchanged for *S. aureus*, and rose by 54 % for *K. pneumoniae*, which is also the pair for which uncertainty acquisition produced the weakest model. We therefore claim a reliable reduction of the dangerous error together with an effect on the other that depends on the organism, rather than a uniform improvement in safety.

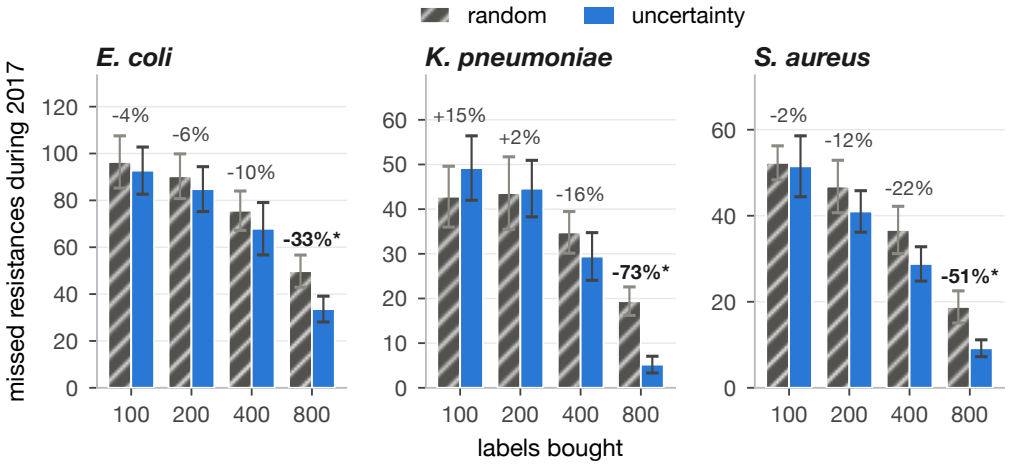


Fig. 6: Resistant isolates wrongly reported susceptible during the year, counted among isolates that were not sent to the laboratory. Bars are means over ten seeds with 95 % confidence intervals; hatched bars are random acquisition and solid bars uncertainty acquisition. Labels give the relative change, with an asterisk marking $p < 0.01$. At small budgets the strategies are indistinguishable; the gap opens as the budget grows.

Table 4: Both error types among untested isolates at a budget of 800 labels, summed over the year, mean over ten seeds. The relative change is uncertainty acquisition against random; p-values are paired Wilcoxon, uncorrected. VME denotes resistant reported susceptible and ME susceptible reported resistant.

Pair	VME random	VME unc.	change	p	ME random	ME unc.	change	p
E. coli	49.8	33.6	−33	0.008	347.8	259.8	−25	0.002
K. pneumoniae	19.4	5.2	−73	0.002	123.7	190.1	+54	0.002
S. aureus	18.8	9.2	−51	0.008	134.6	140.0	+4	0.432

5.4 The Safety Machinery Expires Silently

Both calibrated components failed, and neither failure is visible in the discrimination metric that would normally be monitored. The decision threshold was chosen on 2015 and 2016 validation isolates to catch 90 % of resistant isolates; on 2018 isolates the same threshold caught 80, 73 and 79 % (Fig. 7a). The conformal rule promises that at most 10 % of resistant isolates receive a confident susceptible prediction set; measured over the 2017 isolates it delivered 15.5, 18.9 and 14.3 % (Fig. 7b).

This explains an otherwise puzzling result in E3. Conformal routing matched random acquisition on AUROC while using fewer labels, between 428 and 650 instead of 800, yet left *more* missed resistances than random at the same budget: 78 against 50, 33 against 19 and 39 against 19, with Holm $p < 0.01$ in each case. A routing rule can be no better than the uncertainty estimate it routes on, and that estimate had quietly expired.

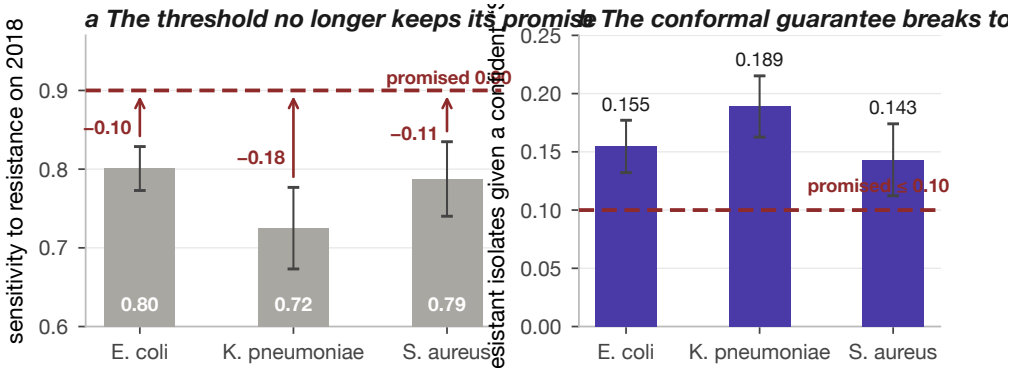


Fig. 7: Two broken promises. (a) The decision threshold was set for 90 % sensitivity to resistance on historical validation data; on 2018 isolates it delivers between 0.10 and 0.18 less. (b) The per class conformal rule promises that at most 10 % of resistant isolates receive a confident susceptible set, and delivers 14 to 19 % over the 2017 isolates. Whiskers are 95 % confidence intervals over seeds.

5.5 Recalibration Restores the Promise and Reveals Its Price

Experiment E5 withholds a quarter of the acquired labels from training and uses them, each round, to reset both the decision threshold and the conformal bounds. Sensitivity on 2018 moves back towards the target, from 0.82 to 0.88, from 0.64 to 0.77 and from 0.72 to 0.79 averaged over strategies and budgets, but the major error rate rises correspondingly, from 0.37 to 0.52, from 0.27 to 0.49 and from 0.26 to 0.39, and balanced accuracy falls (Fig. 8). AUROC is essentially unchanged, marginally lower because a quarter of the acquired labels no longer contributes to training.

The correct reading is not that recalibration is harmful. It is that the drifted threshold had silently moved to a different operating point, one that appeared better on major errors only because it had stopped keeping its sensitivity promise. Restoring the promise reveals what 90 % sensitivity genuinely costs on a model whose AUROC lies between 0.75 and 0.85. A laboratory that finds both numbers unacceptable needs a better model, not a better threshold.

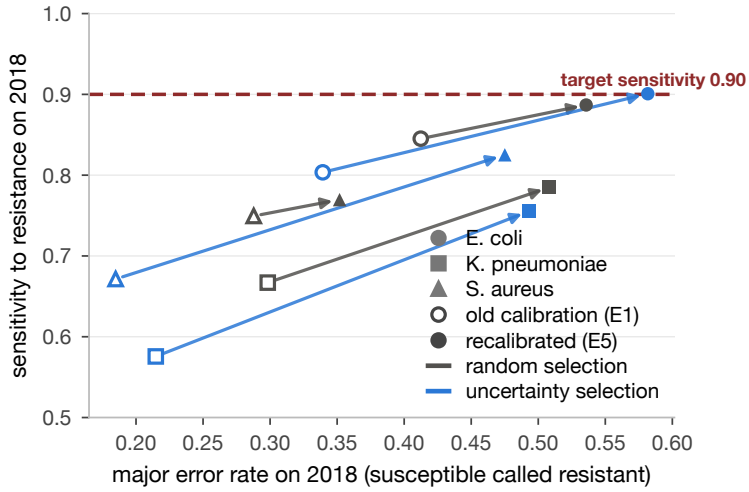


Fig. 8: Recalibration moves every configuration up and to the right: closer to the sensitivity target, at the cost of more susceptible isolates reported resistant. Arrows run from the stale calibration of E1, drawn hollow, to recalibration on freshly acquired labels in E5, drawn filled. Marker shape denotes the species and colour the acquisition rule; each point averages ten seeds and three budgets.

5.6 Adoption at a Different Hospital

Transfer produces the largest effects in the study, and they are not about acquisition. A model trained at hospital A and applied unchanged elsewhere ranged from unaffected, for *E. coli* at hospital B with AUROC 0.835, to worse than chance, for *K. pneumoniae* at hospital B with AUROC 0.477. Local labels repair most of the damage quickly: one hundred locally acquired labels moved that worst case from 0.477 to 0.614, which is the full retraining value at that site (Fig. 9). Which isolates were acquired again made no significant difference in any of 120 comparisons, although the target sites are small and this test has correspondingly little power.

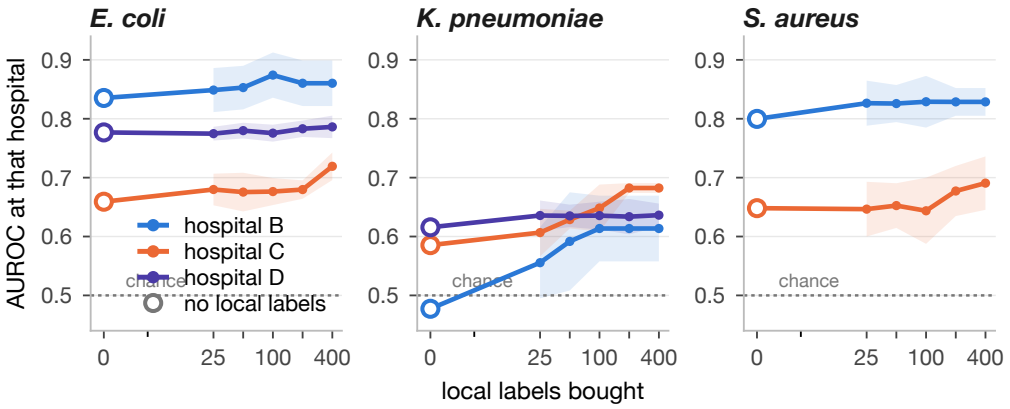


Fig. 9: Adoption of the model at a new hospital. Hollow markers at zero show the transferred model with no local labels; lines show random local acquisition. The collapse for *K. pneumoniae* at hospital B, to below chance, and its repair by 25 to 100 local labels, is the strongest effect observed in the study.

5.7 A Preregistered Search over Method Combinations

A single negative result invites the objection that a better combination would have succeeded. We therefore decomposed the method into five slots, namely acquisition rule, split of the budget between targeted and random purchase, retraining weights, calibration source, and use of unlabelled spectra, then enumerated 2 430 combinations, removed those that are incoherent or dominated, leaving 1 424 valid and 559 both robust and affordable, and selected sixteen to execute. Among them was a family we expected to win: spend half the budget on targeted isolates and half at random, and use the random half, an unbiased sample of the drifted distribution scored before it enters training, to reset the threshold and the conformal bounds.

To keep the search honest, all 510 runs used a **development split that never touches 2018**, training on 2015 and 2016, acquiring labels from the first half of the 2017 arrivals and testing on the second half, and the rule for selecting winners was written into the version controlled log before any run was executed. As a reproducibility check, 110 of the runs were executed independently on two machines and returned identical metrics.

The preregistered rule selected a combination that reduced missed resistances by 63 % and brought sensitivity to 0.92. It achieved this by reporting approximately two thirds of susceptible isolates as resistant. Fig. 10 shows why: every combination that recalibrates slides along the same error tradeoff rather than improving it, and our rule had scored only one end of that tradeoff. We report this as a failure of our own decision rule rather than as a method result, and it is the clearest illustration in this study of how a safety metric read in isolation misleads.

Judged on both error types simultaneously, only one family lies strictly inside the baseline on both axes: uncertainty acquisition, either alone, with a ratio of 0.87 on missed resistances ($p = 0.017$) and 0.84 on major errors ($p = 0.002$) at an AUROC difference of $+0.002$, or mixed half and half with random, with ratios of 0.95 ($p = 0.16$) and 0.89 ($p = 0.006$) at -0.001 AUROC. No combination of recalibration, recency weighting, importance weighting, drift novelty acquisition or conformal pseudo labelling improved on it. Mean AUROC across all sixteen combinations spanned 0.815 to 0.839, so model quality was flat irrespective of the configuration.

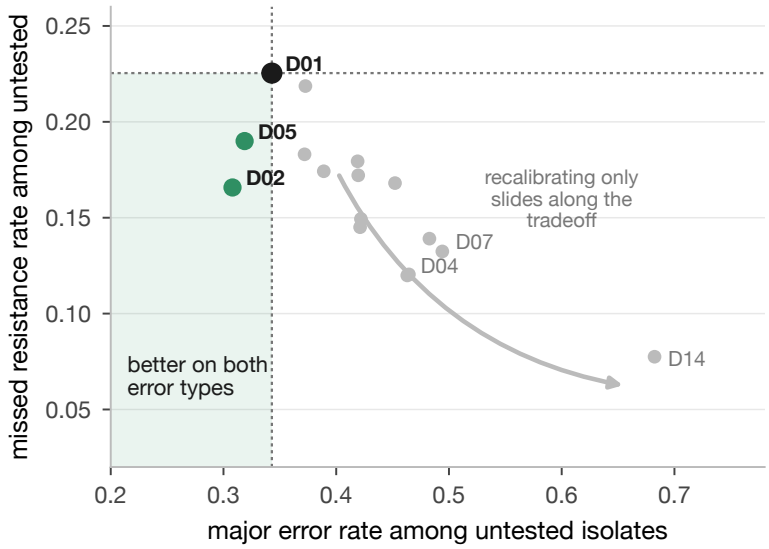


Fig. 10: The error tradeoff on the development split. Each point is one of sixteen method combinations, averaged over three pairs, two budgets and five seeds. Only points in the shaded quadrant improve on the random baseline D01 on both error types. The combination selected by our preregistered rule, D14, lies far to the right: it purchases a low missed resistance rate by reporting most susceptible isolates as resistant.

5.8 An Invalid Experiment: Pseudo labelling

Experiment E2 added self training: isolates about which the model was confident received pseudo labels, with $p \leq 0.05$ treated as susceptible and $p \geq 0.95$ as resistant, and were added to the training set at reduced weight. The results were poor. Pseudo labels alone fell below no retraining at all, and major errors rose sharply at small budgets.

They are also uninterpretable, because of an error in our own design. The decision threshold for these models lies between 0.010 and 0.030, that is, **below the pseudo susceptible cutoff of 0.05**. Isolates in that band were labelled susceptible for training while the deployed model was simultaneously reporting them as resistant (Fig. 11). The procedure therefore injected systematically contradictory labels: between 62 and 66 % of each round's arrivals were pseudo labelled, at 89 to 92 % accuracy, giving 60 to 240 incorrect labels per run. We report the experiment as invalid and leave the semisupervised question open rather than publish a negative result caused by our own implementation. A corrected design ties the cutoffs to the operating threshold or, preferably, derives them from recalibrated conformal sets.

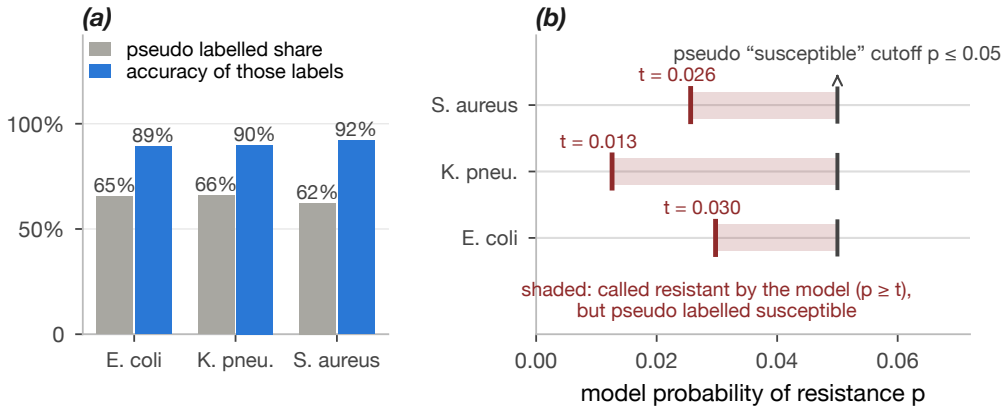


Fig. 11: Why the pseudo labelling experiment is not a fair test. (a) Most arrivals received a pseudo label, roughly one in ten of which was wrong. (b) The fixed cutoff $p \leq 0.05$ for "susceptible" lies above the deployed decision threshold t for all three pairs, so every isolate in the shaded band was trained on as susceptible while being reported as resistant.

6 Discussion

6.1 Label Efficiency Is the Wrong Objective Here

Active learning is conventionally evaluated by how few labels are required to reach a target accuracy. On this task that number is dominated by factors the acquisition rule does not control: the signal available in a 6 000 bin spectrum, the prevalence of the resistant class, and the magnitude of drift between years. Our strategies moved it by less than the variation between seeds. The same choice, meanwhile, moved the number of patients who received a wrong susceptible report by up to a factor of four. For a clinical deployment the second quantity is the one worth optimising, and it is invisible to the standard evaluation.

6.2 Calibration Is a Maintained Component, Not a Training Artefact

Both calibrated components in our system were fitted once and then trusted for two years, which is exactly what a deployed system does by default. Both failed quietly: the threshold lost between 10 and 18 points of sensitivity, and the conformal rule exceeded its own error budget by a factor of 1.4 to 1.9, while AUROC, the number that would ordinarily be monitored, barely moved. Any deployment of this kind of model needs a stream of freshly phenotyped isolates reserved for calibration, and our results suggest an inexpensive way to obtain one: a random share of the testing budget is an unbiased sample of the current distribution and can be scored before it enters training. In our experiments this did not improve the error tradeoff, but it remains worth doing, because it is what keeps a stated guarantee true.

6.3 Practical Recommendations

The findings translate into four concrete recommendations for a laboratory maintaining such a model.

1. Acquire isolates for retraining by whatever means is operationally simplest; random acquisition is not worse in model quality.
2. Where the budget allows, bias acquisition towards isolates about which the model is uncertain, not for the model's benefit, but because those are the isolates whose incorrect calls would otherwise reach patients. Expect the effect on the number of susceptible isolates reported as resistant to vary by organism, and measure it rather than assuming it.
3. Recalibrate the decision threshold and any uncertainty sets on recent labels, and treat a drifting realised sensitivity as an alarm in its own right.
4. When adopting a model developed at another hospital, acquire 25 to 100 local labels before relying on it. This was the single most valuable intervention measured in this study.

7 Limitations

Several limitations qualify these conclusions.

- **No acquisition dates.** DRIAMS does not provide them, so the arrival order within the year is simulated by a hash based partition. Seasonal structure and outbreak dynamics are therefore absent, and the drift within the year that would most favour adaptive acquisition cannot be tested here. This is the limitation we would most like to remove, and it requires a dataset with timestamps.
- **Three pairs, four hospitals, one source.** The organisms with sufficient labelled isolates for this design are few, and all data originate from one country and one collection effort.
- **Simulation rather than deployment.** Labels are revealed from an existing database rather than produced by a laboratory, so turnaround time, batch effects of repeating a test in practice and clinician response are out of scope.
- **Statistical power.** Ten seeds support detection of AUROC differences of approximately 0.008; the transfer tests, with five seeds and small target sites, are weaker. Absence of evidence here is not strong evidence of absence for small effects.
- **One classifier family.** All results use tuned gradient boosted trees. A representation with better behaved predictive uncertainty could change the ranking of acquisition rules, particularly for uncertainty sampling.
- **The semisupervised question remains open** by our own admission, for the reason given in Section 5.8.

8 Conclusion

Across 2 859 retraining simulations on DRIAMS, no active learning strategy produced a better resistance model than acquiring labels at random, and the standard label efficiency endpoint was never reached. The same experiments show that the choice is nevertheless consequential: acquiring the uncertain isolates reduced the resistant isolates wrongly reported susceptible by between a third and three quarters, while the decision threshold and the conformal guarantees that were supposed to bound such errors had both quietly expired after two years of drift. A preregistered search over sixteen method combinations, executed on a development split that never touched the test year, reproduced these conclusions and, by selecting a combination that merely exchanged one error for another, illustrated how readily a safety metric read in isolation misleads. We recommend that acquisition policies for this task be reported with counts at deployment of both error types, and that calibration be treated as a maintained component rather than a property of the training run.

Reproducibility. Every experiment is driven by a single configuration file, and each run writes a self describing record containing its seed, a fingerprint of the data split, library versions and per round metrics. 110 runs were executed independently on two machines and returned identical metrics. Code and result records will be released on publication.

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